

ACT / ALOHA · arXiv 2304.13705 · RSS 2023

Scientific Reviewer

8+3

Cite hygiene · thin real-n
Right cite for chunking + ALOHA

Oct 12 paper club · VLA Architectures

Spine: action chunking as motor contract — TE $\neq$ receding $\neq$ open-loop $\neq$ soft-inpaint

Zhao, Kumar, Levine, Finn · Stanford / Berkeley / Meta

Strengths

Systems + algorithm synergy

Open hardware + ACT; reproducible ethos

Hard fine tasks

Subtask tables; priors often 0% final

Ablations that isolate

k · TE · CVAE with human-vs-scripted contrast

Honest failures

App. F: candy unwrap, flat ziploc, torque limits

Actionable cite hygiene

Spend most of the 8 minutes here

- 1** Real evals 1 seed $\times$ 25 — thin vs sim 3 $\times$ 50; demand CI / more seeds
- 2** BeT/RT-1 without chunks in main tables — gap partly architectural, partly chunking
- 3** $z=0$ at test discards multimodality CVAE was trained for
- 4** Headline 80–90% selective; Insertion sim finals still 32%|20%; Velcro 20%
- 5** No language / multi-task backbone — mark as control+hardware paper
- 6** Cam 30 fps vs control 50 Hz underspecified for visual closed-loop claims

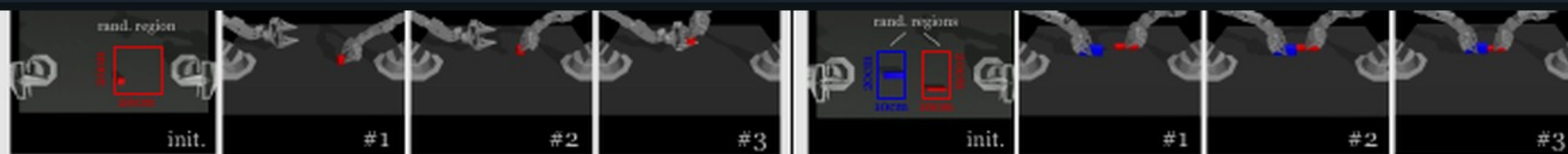
Control schemes — do NOT conflate

Chunking is shared · TE ≠ receding ≠ open-loop ≠ soft-inpaint

Scheme	Mechanism	Commit / loop
ACT TE	Every-step query; exp-avg overlapping preds for same t	ensembled 1-step @ 50 Hz
DP receding	Predict T_p , exec $T_a < T_p$, replan	$T_a \approx 8$ @ ~ 10 Hz
π_0 open-loop	$H=50$; exec prefix; NO TE (hurt early for them)	open-loop prefix @ 20-50 Hz
RTC soft-inpaint	Async; freeze prefix; soft-mask remainder	latency-aware async
BEHAVIOR'25	Predict 30 / exec 26 / keep 4 + correlated FM	receding ~ 30 Hz

Map: $k/50 \leftrightarrow DP\ T_a \cdot \Delta t \leftrightarrow \pi_0/Comet\ H/Hz$ · Defaults: $k=100$ ($\sim 2s$ @ 50 Hz); $L1+\beta KL$; test $z=0$

Table I/II — success rates (sim + real)



Left: Cube Transfer. Transfer the red cube to the other arm. The right arm touches (#1) and grasps (#2) the red cube, then hands it to the left arm.

Right: Bimanual Insertion. Insert the red peg into the blue socket. Both arms grasp (#1), let socket and peg make contact (#2) and insertion.

Fig. 7: *Simulated Task Definitions.* For each of the 2 simulated tasks, we illustrate the initializations and the subtasks.

	Cube Transfer (sim)			Bimanual Insertion (sim)			Slide Ziploc (real)			Slot Battery (real)		
	Touched	Lifted	Transfer	Grasp	Contact	Insert	Grasp	Pinch	Open	Grasp	Place	Insert
BC-ConvMLP	34 3	17 1	1 0	5 0	1 0	1 0	0	0	0	0	0	0
BeT	60 16	51 13	27 1	21 0	4 0	3 0	8	0	0	4	0	0
RT-1	44 4	33 2	2 0	2 0	0 0	1 0	4	0	0	4	0	0
VINN	13 17	9 11	3 0	6 0	1 0	1 0	28	0	0	20	0	0
ACT (Ours)	97 82	90 60	86 50	93 76	90 66	32 20	92	96	88	100	100	96

TABLE I: Success rate (%) for 2 simulated and 2 real-world tasks, comparing our method with 4 baselines. For the two simulated tasks, we report [training with scripted data | training with human data], with 3 seeds and 50 policy evaluations each. For the real-world tasks, we report training with human data, with 1 seed and 25 evaluations. Overall, ACT significantly outperforms previous methods.

	Open Cup (real)			Thread Velcro (real)			Prep Tape (real)				Put On Shoe (real)			
	Tip Over	Grasp	Open Lid	Lift	Grasp	Insert	Grasp	Cut	Handover	Hang	Lift	Insert	Support	Secure
BeT	12	0	0	24	0	0	8	0	0	0	12	0	0	0
ACT (Ours)	100	96	84	92	40	20	96	92	72	64	100	92	92	92

TABLE II: Success rate (%) for the remaining 3 real-world tasks. We only compare with the best performing baseline BeT.

tape dispenser. Especially from the top view, it is challenging to localize the velcro tie because of the small projected area.

the handover, and the left gripper should always move to a position that can grasp the tape, instead of trying to memorize where exactly the handover happens, which can vary across

Real-task finals (ACT) · Table I/II

Slide Ziploc 88% final	Slot Battery 96% final	Open Cup 84% final	Put On Shoe 92% final	Prep Tape 64% final	Thread Velcro 20% final
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Velcro cascade 92→40→20 (perception ceiling) · priors often 0% final

~50 demos/task (100 Velcro) · 1 seed × 25 real · BeT stuck at early stages

Honest: headline 80-90% = easier finals; Insertion sim still hard

Effect-size tattoos · memorize these

Chunking

k=1 → k=100

1% → 44%

sim aggregate, TE off

CVAE (human)

with → without

35.3% → 2%

scripted: ≈ no change

Temporal ensemble

ACT gain

+3.3 pp

BC-ConvMLP +4% · VINN hurt

Teleop Hz

50 Hz vs 5 Hz

~62% slower

user study p<0.001

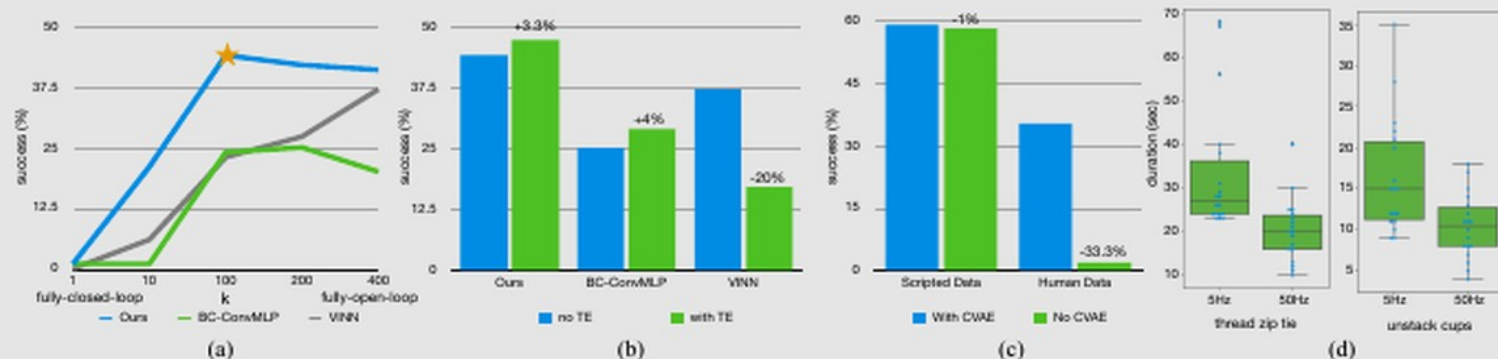


Fig. 8: (a) We augment two baselines with action chunking, with different values of chunk size k on the x-axis, and success rate on the y-axis. Both methods significantly benefit from action chunking, suggesting that it is a generally useful technique. (b) Temporal Ensemble (TE) improves our method and *BC-ConvMLP*, while hurting *VNN*. (c) We compare with and without the CVAE training, showing that it is crucial when learning from human data. (d) We plot the distribution of task completion time in our user study, where we task participants to perform two tasks, at 5Hz or 50Hz teleoperation frequency. Lowering the frequency results in a 62% slowdown in completion time.

(c), we visualize the success rate aggregated across 2 simulated tasks, and separately plot training with scripted data and with human data. We can see that when training on scripted data, the removal of CVAE objective makes almost no difference in performance, because dataset is fully deterministic. While for human data, there is a significant drop from 35.3% to 2%. This illustrates that the CVAE objective is crucial when learning from human demonstrations.

C. Is High-Frequency Necessary?

Lastly, we conduct a user study to illustrate the necessity of high-frequency teleoperation for fine manipulation. With the same hardware setup, we lower the frequency from 50Hz to 5Hz, a control frequency that is similar to recent works that use high-capacity deep networks for imitation learning [7, 70]. We pick two fine-grained tasks: threading a zip cable tie and un-stacking two plastic cups. Both require millimeter-level precision and closed-loop visual feedback. We perform

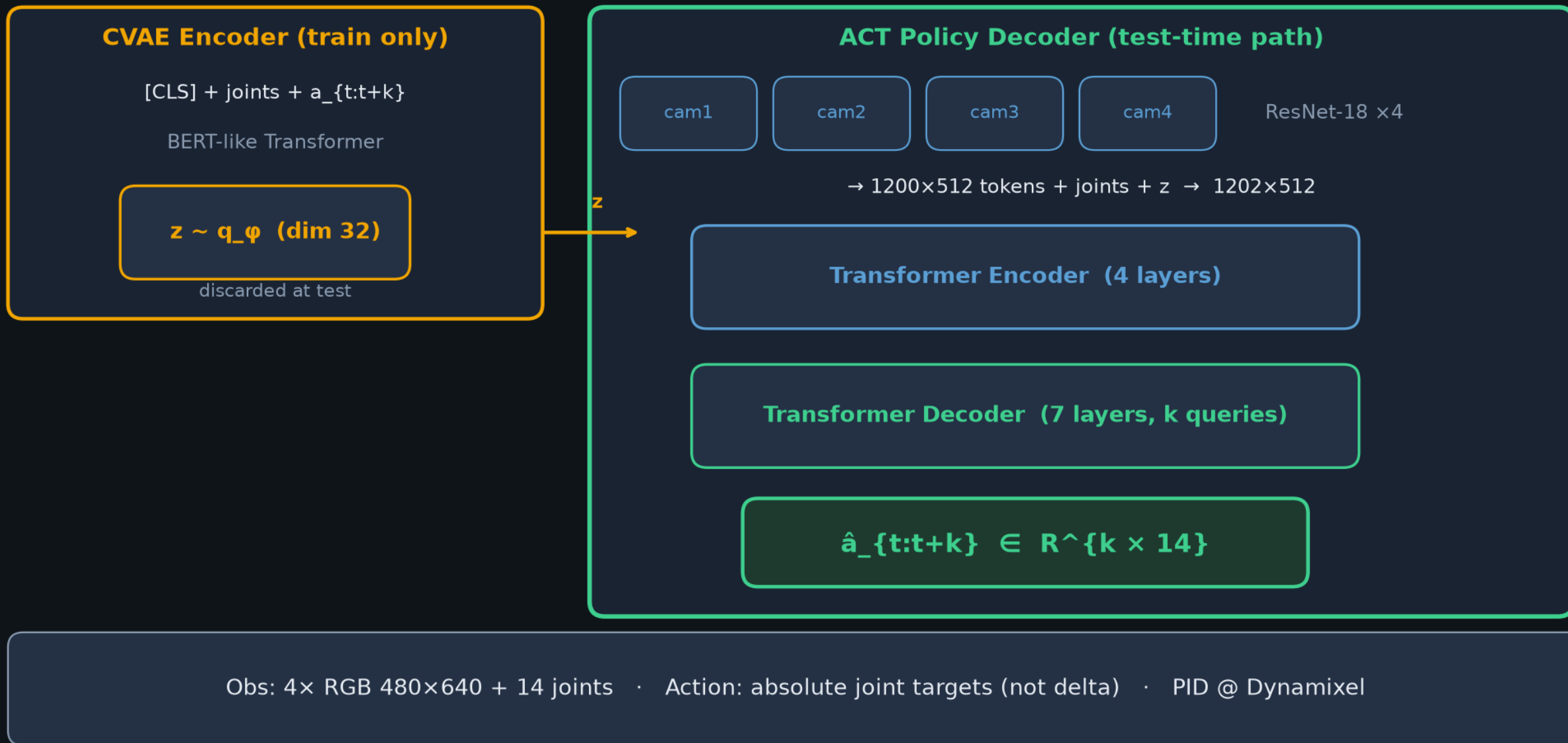
learning algorithm *ACT*. The synergy between these two parts allows us to learn fine manipulation skills directly in the real-world, such as opening a translucent condiment cup and slotting a battery with a 80-90% success rate and around 10 min of demonstrations. While the system is quite capable, there exist tasks that are beyond the capability of either the robots or the learning algorithm, such as buttoning up a dress shirt. We include a more detailed discussion about limitations in Appendix F. Overall, we hope that this low-cost open-source system represents an important step and accessible resource towards advancing fine-grained robotic manipulation.

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ACT architecture · CVAE + Transformer

$\sim 80\text{M} \cdot \text{L1} + \beta \text{KL} (\beta=10) \cdot k \times 14 \text{ absolute joints} \cdot z=0 \text{ at test}$



Verdict

Strong RSS 2023 systems paper

Still the right citation for

action chunking + low-cost bimanual teleop

Scientific core holds · cite Table I/II subtask columns carefully

Right cite · demand hygiene

Cite for action chunking + low-cost bimanual teleop

Flag: real 1×25 · $z=0$ · BeT/RT-1 without chunks in main tables

Q&A · keep the spine clear